

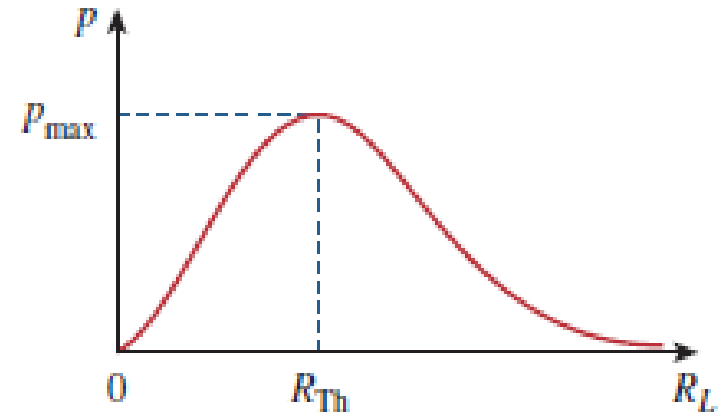
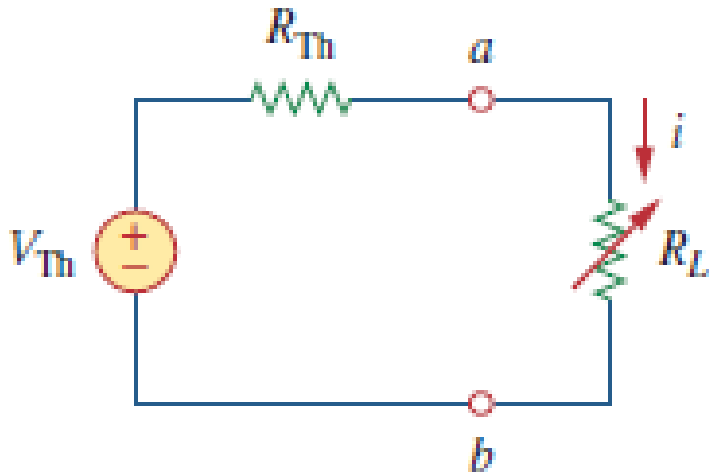
Module 3

Sinusoidal Steady-State Analysis

Dr. Elizabeth Rufus

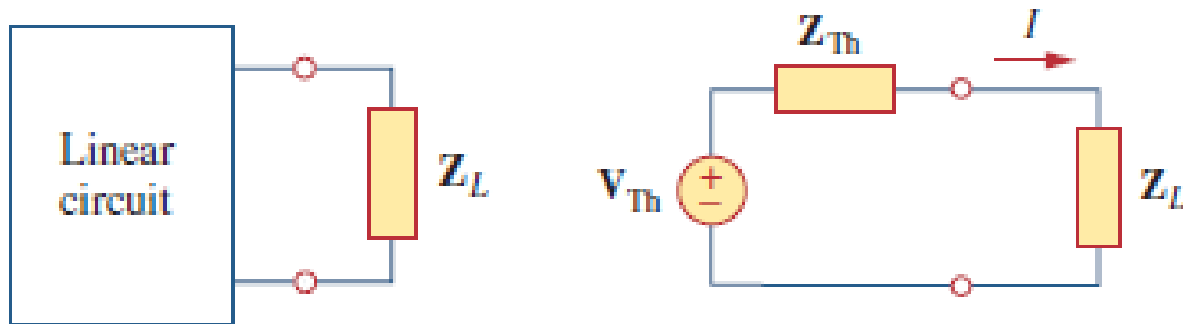
Professor, School of Electronics Engineering

Maximum power transfer theorem



$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

Maximum power transfer theorem



$$Z_{Th} = R_{Th} + jX_{Th}$$

$$Z_L = R_L + jX_L$$

$$Z_L = R_L + jX_L = R_{Th} - jX_{Th} = Z_{Th}^*$$

Setting $R_L = R_{Th}$ and $X_L = -X_{Th}$

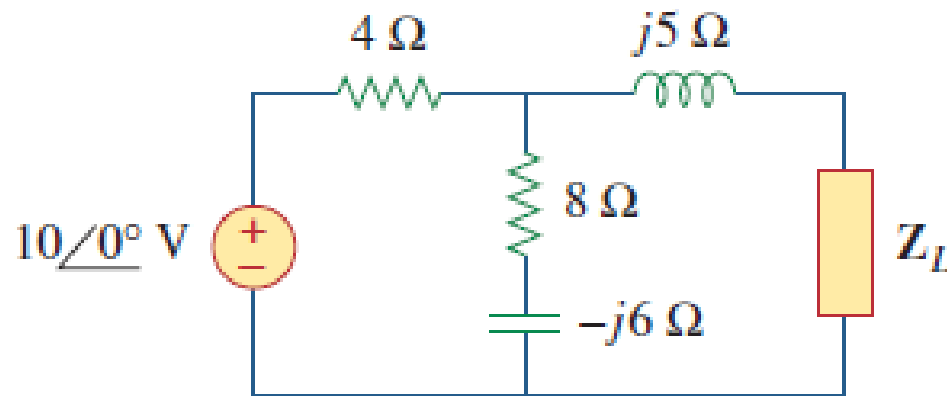
$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}}$$

$$R_L = \sqrt{R_{Th}^2 + X_{Th}^2} = |Z_{Th}|$$

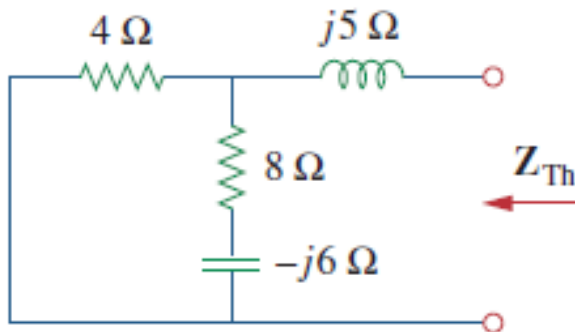
Maximum power transfer theorem

1. Determine the load impedance Z_L that maximizes the average power drawn from the circuit.

What is the maximum average power.



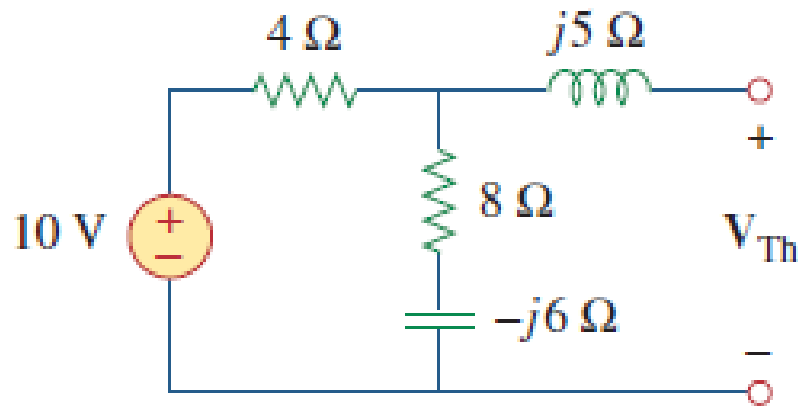
Find Z_{Th} ,



$$\begin{aligned} Z_{Th} &= j5 + 4 \parallel (8 - j6) = j5 + \frac{4(8 - j6)}{4 + 8 - j6} \\ &= 2.933 + j4.467 \, \Omega \end{aligned}$$

Maximum power transfer theorem

Find V_{Th} ,



$$V_{Th} = \frac{8 - j6}{4 + 8 - j6}(10) = 7.454 \angle -10.3^\circ \text{ V}$$

The load impedance draws the maximum power from the circuit when

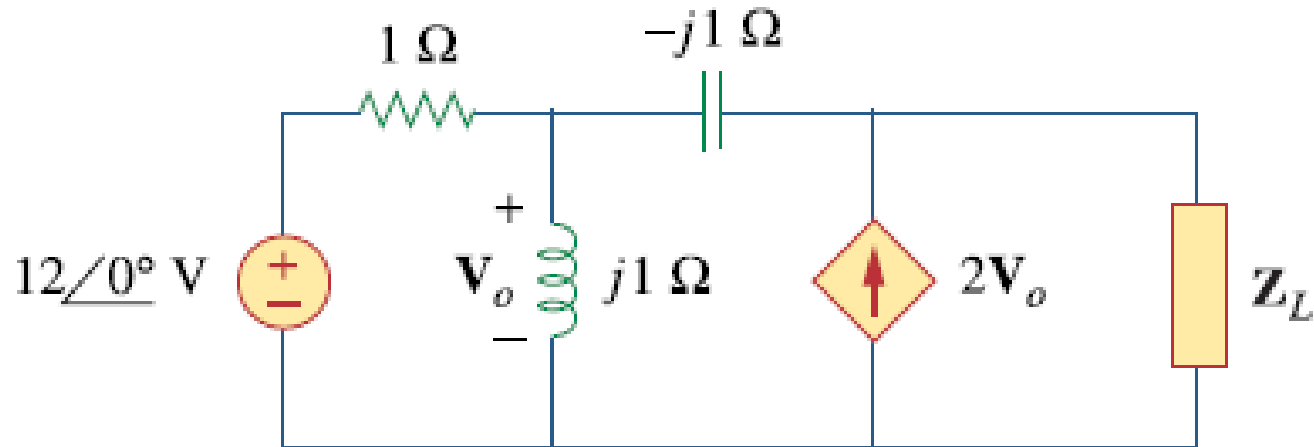
$$Z_L = Z_{Th}^* = 2.933 - j4.467 \Omega$$

the maximum average power is

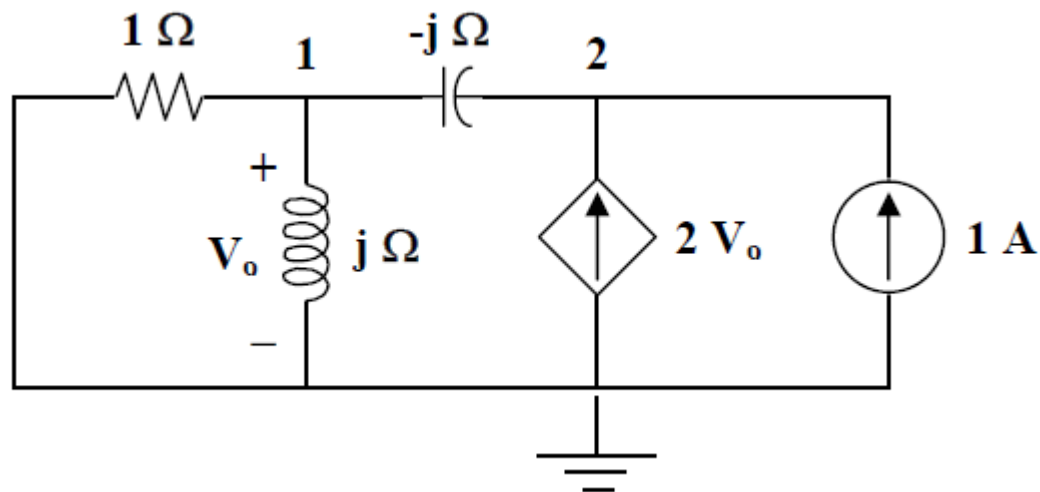
$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}} = \frac{(7.454)^2}{8(2.933)} = 2.368 \text{ W}$$

Maximum power transfer theorem

2. Find the value of Z_L that will absorb the maximum power and the value of the maximum power.



To find Z_{Th} , insert a 1-A current source at the load terminals



Maximum power transfer theorem

At node 1,

$$\frac{\mathbf{V}_o}{1} + \frac{\mathbf{V}_o}{j} = \frac{\mathbf{V}_2 - \mathbf{V}_o}{-j} \longrightarrow \mathbf{V}_o = j\mathbf{V}_2 \quad (1)$$

At node 2,

$$1 + 2\mathbf{V}_o = \frac{\mathbf{V}_2 - \mathbf{V}_o}{-j} \longrightarrow 1 = j\mathbf{V}_2 - (2 + j)\mathbf{V}_o \quad (2)$$

Substituting (1) into (2),

$$1 = j\mathbf{V}_2 - (2 + j)(j)\mathbf{V}_2 = (1 - j)\mathbf{V}_2$$

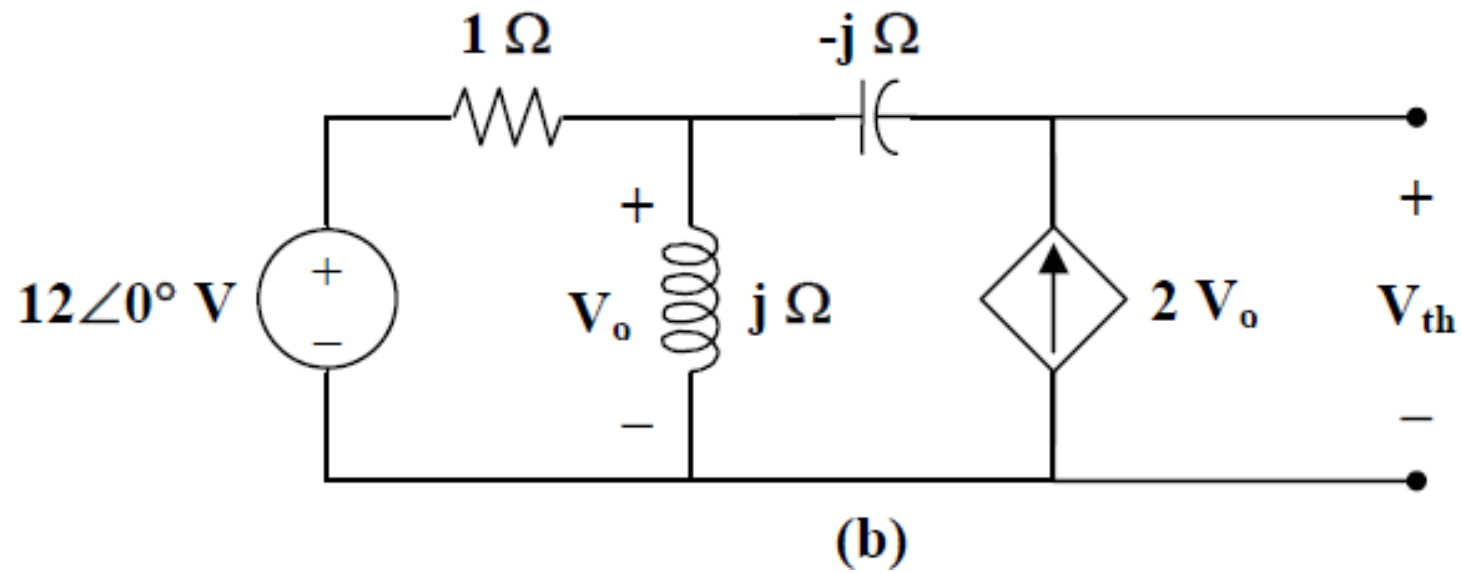
$$\mathbf{V}_2 = \frac{1}{1 - j}$$

$$\mathbf{Z}_{Th} = \frac{\mathbf{V}_2}{1} = \frac{1 + j}{2} = 0.5 + j0.5$$

$$\mathbf{Z}_L = \mathbf{Z}_{Th}^* = \underline{\underline{0.5 - j0.5 \Omega}}$$

Maximum power transfer theorem

We now obtain V_{Th} from Fig. (b).



Maximum power transfer theorem

$$2\mathbf{V}_o + \frac{12 - \mathbf{V}_o}{1} = \frac{\mathbf{V}_o}{j}$$

$$\mathbf{V}_o = \frac{-12}{1 + j}$$

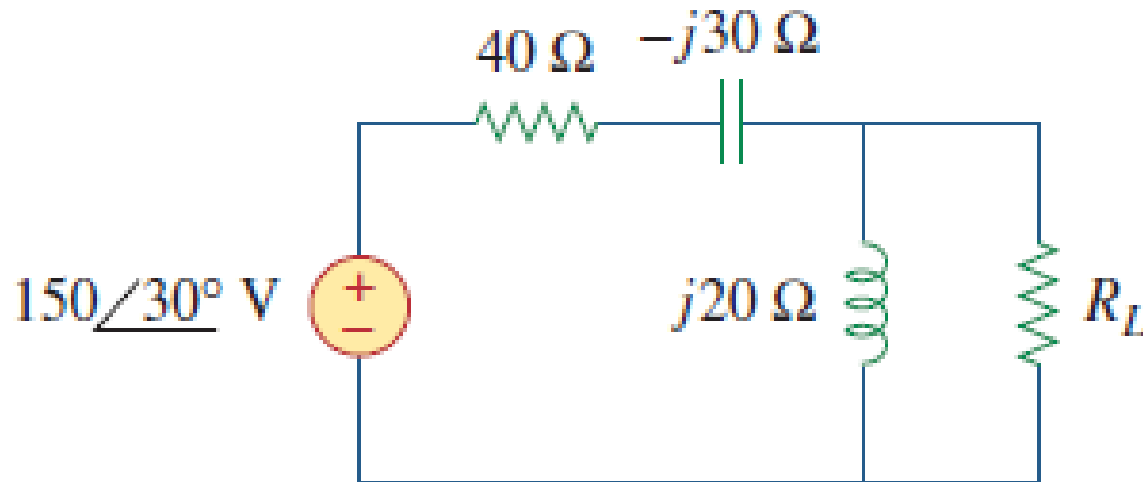
$$-\mathbf{V}_o - (-j \times 2\mathbf{V}_o) + \mathbf{V}_{Th} = 0$$

$$\mathbf{V}_{Th} = (1 - j2)\mathbf{V}_o = \frac{(-12)(1 - j2)}{1 + j}$$

$$P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_L} = \frac{\left(\frac{12\sqrt{5}}{\sqrt{2}}\right)^2}{(8)(0.5)} = \underline{\underline{90 \text{ W}}}$$

Maximum power transfer theorem

3. Find the value of R_L that will absorb the maximum average power. Calculate the power.



$$Z_{\text{Th}} = (40 - j30) \parallel j20 = \frac{j20(40 - j30)}{j20 + 40 - j30} = 9.412 + j22.35\ \Omega$$

By voltage division,
$$V_{\text{Th}} = \frac{j20}{j20 + 40 - j30} (150\angle 30^\circ) = 72.76\angle 134^\circ \text{ V}$$

Maximum power transfer theorem

The value of R_L that will absorb the maximum average power is

$$R_L = |Z_{Th}| = \sqrt{9.412^2 + 22.35^2} = 24.25 \, \Omega$$

The current through the load is

$$\mathbf{I} = \frac{\mathbf{V}_{Th}}{\mathbf{Z}_{Th} + R_L} = \frac{72.76 \angle 134^\circ}{33.66 + j22.35} = 1.8 \angle 100.42^\circ \, \text{A}$$

The maximum average power absorbed by R_L is

$$P_{\max} = \frac{1}{2} |\mathbf{I}|^2 R_L = \frac{1}{2} (1.8)^2 (24.25) = 39.29 \, \text{W}$$

Next class: Module 4